MIL-L-48385 (PA)
03 March 1975

MILITARY SPECIFICATION

LINE, EXTENSION AND SPOOL ASSEMBLY
FOR MINE, APERS, M18A1E1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the assembling and packing for one type Line, Extension and Spool Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes (ABC-STD-105)
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1345

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

9271910 - Extension Line and Spool Assembly

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Extension Line and Spool Assembly. - The extension line and spool assembly shall comply with all requirements specified on Drawing (dwg.) 9271910 and with all requirements specified in applicable specifications.

3.3 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

3.4 First article inspection. - This specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample (see 6.2b) as designated by the Contracting Officer. The first article sample shall consist of 10 complete sets of parts and sub-assemblies (i.e., 10 each of every component and sub-assembly, including at least five (5) items from each cavity used to produce molded parts, and be so identified as to permit tracing each item to the cavity in which it was molded). The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

4.3.2 Inspections to be performed. - See MIL-A-48078 and paras. 4.4.2, testing per para. 4.4.3 and molding inspections per para. 4.4.5.

4.3.3 Rejection. - (See MIL-A-48078)

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Lot numbering as required shall be in accordance with MIL-STD-1168.

4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-48385 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8861595
4.4.2.1	Shorting Prong				NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Length		0.40%	3.2	Gage
102.	Diameter of ends		0.40%	3.2	Gage
103.	Distance between centers of ends		0.40%	3.2	Gage
Minor:					
201.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

SARPA-QA FOR 367 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Prong				9268999
					NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Total length		0.40%	3.2	Gage
102.	Prong end length		0.40%	3.2	Gage
103.	Prong end diameter		0.40%	3.2	Gage
Minor:					
201.	Knurl missing or incomplete		0.65%	3.2	Visual
202.	Hole depth		0.65%	3.2	Gage
203.	Body length		0.65%	3.2	Gage
204.	Hole diameter		0.65%	3.2	Gage
205.	Shank diameter		0.65%	3.2	Gage
206.	Blow hole missing		0.65%	3.2	Visual
207.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

5

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9269002
4.4.2.3	Female contact, prior to assembly of spring insert				NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Total length		0.40%	3.2	Gage
102.	Prong hole length		0.40%	3.2	Gage
103.	Diameter of prong hole		0.40%	3.2	Gage
Minor:					
201.	Knurl missing or incomplete		0.65%	3.2	Visual
202.	Wire hole diameter		0.65%	3.2	Gage
203.	Wire hole depth		0.65%	3.2	Gage
204.	Body length		0.65%	3.2	Gage
205.	Shank diameter		0.65%	3.2	Gage
206.	Blow hole missing		0.65%	3.2	Visual
207.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS MIL-L-48385 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Female contact				9269002
					NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Spring insert missing		0.40%	3.2	Visual
102.	Swage not 360 degrees		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

1

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

8

CLASSIFICATION OF DEFECTS & TESTS MIL-L-48385 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Extension line (prior to assembly on spool)				9269004
					NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Insulation damaged or enough insulation missing to result in shorting		0.40%	3.2	Visual
Major B:					
131.	Length of cable (exclusive of connectors)		0.40%	3.2	Scale
Minor:					
201.	Color of wire incorrect		0.65%	3.2	Visual
202.	Excessive mold flash on male and female connectors		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

SARPA-QA FORM 1367 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9271910
4.4.2.6	Extension line and spool assembly				NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:	None defined				
Minor:					
201.	Carrying cord short (prior to tying)		0.65%	3.2	Scale
202.	Carrying cord missing		0.65%	3.2	Visual
203.	Connectors not securely taped		0.65%	3.2	Visual/Manual
204.	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
205.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

\o

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Line, Extension (prior to molding)				9269004
					NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Bare wire showing between cable and female contact over max. (prior to crimping contact and molding in connector)		0.40%	3.2	Visual
132.	Bare wire showing between cable and prong over max. (prior to molding in connector)		0.40%	3.2	Visual
133.	Incorrect trim at wire ends		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Pull test of crimped connections. - (See dwg. 9269004), Major A defect. Fifty (50) connectors shall be selected from each lot for test. The lot shall be rejected if any connector fails to comply with the requirements. The test shall be performed as specified in 4.5.1 using test equipment in accordance with 4.4.4. (Non-destructive)

4.4.3.2 Resistance of extension line. - (see dwg. 9269004) Major A defect. This test shall be performed 100 percent as specified in 4.5.2 using equipment in accordance with 4.4.4. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. (Non-destructive test).

4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed In this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 (PA).

4.4.5 Dimensional control of molded parts. - (The requirements of this paragraph apply to those dimensions which are considered "as molded", and which do not undergo subsequent operations). In place of normal sampling and inspections associated with the Classification of Defects, and with the arpoval of the Contracting Officer, and after a curing time for the parts has been approved (see Note 1) a sample of five (5) parts (as molded) from each cavity shall be inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification, when more than one cavity is used in production. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.5.1 through 4.4.5.3 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the five (5) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of produc- for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1. - In establishing the curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey 07801 ATTN: SARPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.5.1	Line, Extension (female connector)				9269004
					NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Presense of rubber or other material in the contact holes	5	100%	3.2	Visual
Major B:					
131.	Depth of top end of female contacts	5	100%	3.2	Gage
132.	True position of female contacts	5	100%	3.2	Gage
133.	True position of shorting prong	5	100%	3.2	Gage
134.	Length of connector	5	100%	3.2	Gage
135.	Recessed depth to outer end of prong	5	100%	3.2	Gage
136.	Recessed depth to dust cap	5	100%	3.2	Gage
Minor:					
201.	Dust cap does not fit securely	5	100%	3.2	Visual/Manual
202.	Sleeve missing	5	100%	3.2	Visual
203.	Evidence of poor workmanship	Sample	0.65%	3.3	Visual

NOTES:

1

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS MIL-L-48385 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9269004
4.4.5.2	Line, Extension (male connector)				NEXT HIGHER ASSEMBLY 9271910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Presence of rubber or other foreign material on exposed surfaces of prongs	5	100%	3.2	Visual
Major B:					
131.	True position of prongs	5	100%	3.2	Gage
132.	Recessed depth of connector	5	100%	3.2	Gage
133.	Width of connector recess, min.	5	100%	3.2	Gage
134.	Length of connector	5	100%	3.2	Gage
Minor:					
201.	Prong shoulders not flush with bttom surface	5	100%	3.2	Visual
202.	Dust cap does not fit securely	5	100%	3.2	Visual/Manual
203.	True position of dust cap holes	5	100%	3.2	Gage
204.	Diameter of dust cap holes	5	100%	3.2	Gage
205.	Evidence of poor workmanship	Sample	0.65%	3.3	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9271771 / NEXT HIGHER ASSEMBLY 9271910
4.4.5.3	Spool				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Crack or split	5	100%	3.2	Visual
Minor:					
201.	Length of spool	5	100%	3.2	Gage
202.	Diameter of flange	5	100%	3.2	Gage
203.	Thickness of flange	5	100%	3.2	Gage
204.	Notch in flange missing	5	100%	3.2	Visual
205.	Diameter of core	5	100%	3.2	Gage
206.	Contour of notch incorrect	5	100%	3.2	Gage
207.	Evidence of poor workmanship	Sample	0.65%	3.3	Visual

NOTES:
15

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

4.5 Test methods and procedures

4.5.1 Pull test of crimped connections - Before molding into connector assemblies, the cable (wire) shall be held securely and the required force applied to each contact (female and prong) at a rate not to exceed 1/8 inch per minute. Observation shall be made for compliance with the applicable requirements.

4.5.2 Resistance of extension line. - This test shall be performed after molding of connectors to cable (wire). The resistance shall be measured at the male connector with female connector shorted. Observation shall be made for compliance with the applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Marking - Level C. - The assemblies shall be packaged, packed and marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended Use. - The extension line and spool assembly Is intended to be used with the M18A1E1 Mine.

6.2 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval - (See MIL-A-48078). Submit equipment designs as required to Commander: Picatinny Arsenal, ATTN: SARPA-A-QA-T- Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M Dover, New Jersey 07801.

Custodian:
Army-PA

Preparing Activity:
Army - PA

Project Number: 1345-A-085

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.